Lab 7: Fuzzing for Input Validation Bugs - PES1UG24CS498

Step 2: Output of the original (buggy) processor.py

```
PS C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy> python processor.py
==== Running parse_transit_corridor ====
Enter corridor (e.g. DT-UP-3): bad
Traceback (most recent call last):
  File "C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy\processor.py", line 39, in <module>
    main()
    ~~~~^^
  File "C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy\processor.py", line 28, in main
    print(parse_transit_corridor(c_input))
          ~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^
  File "C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy\processor.py", line 2, in parse_transit_corridor
    orig, dest, zones_str = corridor_code.strip().split('-')
    ^^^^^^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 3, got 1)
```

Step 4: pytest test_processor.py against the buggy processor (tests failing)

```
C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy> python -m pytest test_processor.py
  +---------------- 3 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy\test_processor.py", line 21, in test_compute_passenger_density_safe
  |     compute_passenger_density(s)
  |     ~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
  |   File "C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy\processor.py", line 19, in compute_passenger_density
  |     passengers_str, km_str = occupancy_ratio_str.split('/')
  |                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'split'
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\TEJAS PONNAPPA\Downloads\lab7_buggy\test_processor.py", line 23, in test_compute_passenger_density_safe
  |     assert False, f"compute_passenger_density crashed with: {e!r}"
  | AssertionError: compute_passenger_density crashed with: AttributeError("'NoneType' object has no attribute 'split'")
  | assert False
  | Failing test case: test_compute_passenger_density_safe(
  |     s=None,  # or any other generated value
  | )
  +-----------------------------------
=========================================================== short test summary info ============================================================
ILED test_processor.py::test_parse_transit_corridor_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
ILED test_processor.py::test_lookup_concession_discount_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
ILED test_processor.py::test_compute_passenger_density_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
```

```
=========================================================== short test summary info ============================================================
FAILED test_processor.py::test_parse_transit_corridor_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_lookup_concession_discount_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_compute_passenger_density_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================================================ 3 failed in 3.49s =============================================================
```

Step 5: Output of the fixed processor.py

```
==== Running parse_transit_corridor ====
Enter corridor (e.g. DT-UP-3): bad
Invalid corridor code
```

Step 6: pytest test_processor.py against the fixed processor (tests passing)

